

Bonus Project: Deep Learning (SS26)

Multivariate Time Series Forecasting (Exposé)



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1 Experimental Setup

1.1 Intended Model Family

To address the multivariate time series forecasting task, we propose using a Recurrent Neural Network (RNN), specifically a Long Short-Term Memory (LSTM) network [1]. LSTMs are a well-established architecture for modeling sequential data and capturing long-term temporal dependencies. According to Mienye et al., RNNs are designed to learn sequential and time-varying patterns, making them particularly suitable for time series applications [2]. RNN-based models have been successfully applied to a wide range of forecasting problems, including electric load forecasting [3] and financial prediction [4]. Since the dataset used in this project consists of multivariate weather-related time series data, an LSTM-based approach is a natural choice. The selected architecture aligns with proven methods for modeling temporal dependencies and generating accurate forecasts from sequential data.

1.2 Baselines

For a quick evaluation of the models performance we may use the provided baseline generator. When looking closely at the data a boolean feature called *is_weekend* can be obtained easily. With small effort the baseline results might be increased by distinguishing with that feature, f.e. by using the last 1, 24 or 168 values within the respective value for *is_weekend*.

2 Additional Dataset Experiment

For multivariate time series forecasting on a further dataset we decided to use a weather dataset from the *Max-Planck Institute for Biogeochemistry* [5]. It can be considered as somewhat famous in the Machine Learning world as it has been studied by researchers all around the world. It contains weather data between 2009 and 2016 and is provided to public via kaggle [6]. In the dataset every 20 minutes a new entry with 14 features was stored. The resulting dataset consists of 421.000 entries and the csv-document has a size of 41.99 MB. There are several download sources online but they likely refer on the almost identical dataset.

2.1 Experimental Setup

The dataset includes 14 features which is why we can more or less use the same architecture and model family as for the given datasets.

The given dataset is not splitted into training, validation and test subsets, making it crucial for us to do that manually. Initial considerations suggest a split of f.e. 85%, 10% and 5%. In order to prevent overfitting early stopping the adaptation of early stopping might increase the performance of our model.

2.2 Key Risks And Open Questions

The precipitation is measured with tipping bucket rain gauges which may cause some variable deviations between two stations. This kind of error is a so called wind error.

Furthermore a blockage due to dirt inside the instrument which has not been removed may cause another error in calculation. The rain gauges are heated.

In case of snowfall or hail events it can get detached. This results in a delayed recording to the water equivalent [5].

3 Baselines

One simple idea for having a working baseline for our weather forecasting dataset is using and slightly modifying the baseline generators provided for this exercise. They should provide a quick solution, but we expect a more or less modest performance. But as the dataset is provided by kaggle and a few other people already published a solution for the dataset, this gives us the opportunity to compare our results with other techniques and concepts. One interesting solution is by Koushik M. and uses AR, MA and ARMA models, among other non-stationary models and also Deep Learning concepts [7]. Koushik even implemented some interesting baseline generators by using the values from the same time but one day or one week before. These are easy to implement and should give us a good starting point for evaluating the performance of our Deep Learning model.

References

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